Q-1 What comes next in the following series –
6, 1, 6, 2, 6, 4, 6, 8, 6, _____ ?

A 10
B 12
C 14
D 16

Correct Answer : D

Q-2 FLOCK : BIRDS :: SWARM : ___?

A BEES
B FISH
C BATS
D SHEEP

Correct Answer : A

Q-3 What comes next in the following series –
AN, CP, ER, ___?

A AD
B GT
C ST
D DT

Correct Answer : B

Q-4 Find the odd one out in the given series: Violin, Guitar, Flute, Sitar.

A Violin
B Guitar
C Flute
D Sitar

Correct Answer : C

Q-5 Find odd one out from the given list of jumbled words: LNUAIMMIU, ELEST, OPERPC, NMKOYE, INCKLE

A ELEST
B OPERPC
C NMKOYE
D INCKLE

Correct Answer : C

Q-6 What is the Cartesian product of A = {1, 4} and B = {a, b}?

A {(1,a), (4,a), (1,b), (4,b)}
B {(1,1), (4,4), (a,a), (b,b)}
C {(1,a), (1,b), (4,a), (b,b)}
D {(1,1), (a,a), (4,a), (1,b)}

Correct Answer : A

Q-7 Which of the following is not an Operating system?

A Ubuntu
B Lubuntu
C Windows
D Solabis

Correct Answer : D

Q-8 Which of the following communication mediums is not used for digital communication?

A Shielded twisted pair cables
B Optical fiber
C Coaxial cable
D Un-shielded twisted pair cables

Correct Answer : B

Q-9 Which of the following addresses is a physical address of a computer?

A **MAC address**
B IP address
C Email address
D IMEI number

Correct Answer : A

Q-10 Which of the following addresses is used for routing information in a network?

A MAC address
B Gateway MAC address
C **IP address**
D Application address

Correct Answer : C

Q-11 Which of the following modules performs the arithmetic operations in a computer?

A **ALU**
B CMU
C RAM
D Accumulator

Correct Answer : A

Q-12 What is the size of an IPv6 address?

A 32 bits
B 64 bits
C **128 bits**
D 256 bits

Correct Answer : C

Q-13 Which of the following frequency bands is used for Wi-Fi networks?

A 100MHz
B 99.1MHz
C 2GHz
D **5GHz**

Correct Answer : D

Q-14 Which of the following components is NOT typically found on a computer motherboard?

A CPU socket
B RAM slots
C Graphics card
D **Power supply unit**

Correct Answer : D

Q-15 Which one out of the following is not a flowchart symbol?

A Circle
B Rectangle
C **Triangle**
D Diamond

Correct Answer : C

Q-16 Identify the variables in the following list:
'x', "x", x, 5.

A 'x', "x", x
B 'x', "x"
C 'x', x
D **x**

Correct Answer : D

Q-17 Fill in the blank with the correct preposition: "She is good _____ numbers."

A in
B on
C by
D **at**

Correct Answer : D

Q-18 What is the opposite of "lazy"?

A Sad

B Excited

C Hardworking

D Sleepy

Correct Answer : C

Q-19 What does the word "enormous" mean?

A midget

B small

C infinitesimal

D Colossal

Correct Answer : D

Q-20 Choose the correct sentence.

A By the time we arrived, the show has started.

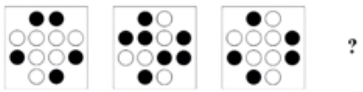
B By the time we arrived, the show was starting.

C By the time we arrived, the show had started.

D By the time we arrived, the show starts.

Correct Answer : C

Q-21 What comes next in the following series of geometric figures?



A

B

C

D

Correct Answer : A

Q-22 In a certain encoding, the word TIER is written as 7163 and the word BRAIN is written as 23415. What will be the encoding for the word RENT?

A 6315

B 3567

C 4153

D 3657

Correct Answer : D

Q-23 If "BLUE" is given the number code 40, and "RED" is given the code 27, what is the difference between the number codes of "PINK" and "GREEN" ?

A 13

B 10

C 4

D 1

Correct Answer : D

Q-24 Rashmi and Rajesh apply for two vacant spots in a company. If the probability of Rajesh getting selected and Rashmi getting selected are 3/7 and 2/3 respectively, what is the probability that neither of them will be selected?

A 2/7

B 6/21

C 5/21

D 4/21

Correct Answer : D

Q-25 Let A={1,2} and B={3,4}. Which of the following cannot be relation from set A to set B?

A {(1,3), (1,4)}

B {(2,3), (2,4)}

C {(1,1), (1,2), (1,3), (1,4)}

D {(1,3), (1,4), (2,3), (2,4)}

Correct Answer : C

Q-26 Let A={1,2,3} and B={4,5,6}. Which one of the following functions is bijective?

A f={(2,4),(2,5),(2,6)}

B f={(1,5),(2,4),(3,4)}

C f={(1,4),(1,5),(1,6)}

D f={(1,4),(2,5),(3,6)}

Correct Answer : D

Q-27 Find the number of ways of arranging the letters of the words GRADER, so that no vowel occupies odd place.

A 36

B 48

C 144

D 96

Correct Answer : C

Q-28 Which of the following is the primary reason for using a hash table in computer science?

A To store elements in sorted order

B To allow for efficient searching and insertion

C To store elements that are linked together

D To provide constant time complexity for sorting

Correct Answer : B

Q-29 Which of the following is an advantage of using a stack over a queue?

A It allows for FIFO (First-In-First-Out) operations

B It allows for LIFO (Last-In-First-Out) operations, useful in recursive algorithms

C It allows random access to elements

D It supports efficient enqueue and dequeue operations from both ends

Correct Answer : B

Q-30 When dealing with floating-point arithmetic, which of the following is a common source of error?

A Integer overflow

B Rounding errors

C Stack overflow

D Type mismatch errors

Correct Answer : B

Q-31 What is the final value of x printed?

```
int x = 1;
if (x && --x){
   x+=2;
}
printf("%d\n",x);
```

A 0

B 1

C 5

D 4

Correct Answer : A

Q-32 Given q=20, r=30, s=40, what will be the value of the integer X after execution of the C statement, X = ++q * r / s ;

A 15.75

B 16.75

C 16

D 15

Correct Answer : D

Q-33 What will be the value of x after executing the following statements?

```
x=8; y=5;
x=x+y;
y=x-y;
x=x-y;
```

A 13

B 3

C 5

D 8

Correct Answer : C

Q-34 Choose the correct word: "The instructor asked us to _____ the paragraph."

A summerize

B summarise

C summarisse

D summirize

Correct Answer : B

Q-35 Select the correct sentence.

A The dog chased the ball.

B The dogs chased its ball.

C The dog chase it's ball.

D The dog chase its tail.

Correct Answer : A

Q-36 If Rima says, "Tara's father Raman is the only son of my father-in-law Ramanand", then how is Sitara, who is the sister of Tara, related to Ramanand?

A Daughter

B Grand daughter

C Niece

D Sister

Correct Answer : B

Q-37 Ravi was on his way home when a boy asked him to donate some money to the poor people's fund. Ravi gave him one rupee more than half the money he had. He walked a few more steps when he met a girl who requested him to donate for the animal welfare society. This time he gave two rupees more than half the money he had then. After some time, he met another girl who requested him to donate for an orphanage. He gave three rupees more than half of what he had then and was left with only one rupee. How much money did Ravi have with him when he started?

A Rs. 42

B Rs.52

C Rs.62

D Rs.72

Correct Answer : A

Q-38 There are 350 students who appeared in an examination, out of them only 48% passed in mathematics, 46% passed in science and 38% failed in both subjects. How many of these students passed in both the subjects?

A 135

B 175

C 112

D 196

Correct Answer : C

Q-39 Let A and B be two events such that the occurrence of A implies occurrence of B, but not vice-versa. Then the correct relation between the probabilities P(A) and P(B) is _____.

A $P(A) < P(B)$

B $P(B) \geq P(A)$

C $P(A) = P(B)$

D $P(A) \geq P(B)$

Correct Answer : B

Q-40 The maximum number of equivalence relations on the set A = {1, 2, 3} is _____.

A 1

B 2

C 3

D 5

Correct Answer : D

Q-41 The equation $(a + 1)^2 - 2(a + 1) = 0$ has _____.

A **two real roots**
B two equal roots
C one real root
D no real roots

Correct Answer : A

Q-42 At a certain university, 9% of men are over 6 feet tall and 4% of women are over 6 feet tall. The total student population is divided in the ratio 3:4 in favour of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman?

A 3/7
B 4/7
C **16/43**
D 1/100

Correct Answer : C

Q-43 What is the output of the following C++ program?

```
#include <iostream>
using namespace std;

void modify(int &a, int b) {
  a = a + b;
  b = a - b;
}

int main() {
  int x = 5, y = 10;
  modify(x, y);
  modify(x,y);
  cout<<"x= "<<x<<"y="<<y<<endl;
}
```

A x=15 y=5
B x=15 y=10
C x=25 y=15
D **x=25 y=10**

Correct Answer : D

Q-44 What is the output of the following C program?

```
#include <stdio.h>
#define sqr1(x) x * x
#define sqr2(x) (x) * (x)
int main() {
   int a = 6;
   printf("%d %d\n", sqr1(a + 1), sqr2(a+1));
   return 0;
}
```

A **13 49**
B 36 49
C 49 36
D 37 49

Correct Answer : A

Q-45 Find the value of z after executing the following code fragment?

```
z=0;
for ( x = -1 to 1 step 1)
   for(y = x+1 to  5 step 2)
         z=z+y;
```

A 12
B **21**
C 24
D 33

Correct Answer : B

Q-46 Find the value of m after the following statements are executed?

```
n=235;
m=0;
while ( n > 0)
   {m = m*10 + n MOD 10;
    n = int(n/10);
   }
```

A **532**

B 32

C 35

D 235

Correct Answer : A

Q-47 Identify the adjective in the following sentence:_x000D_
"She was the most well-dressed lady in the party."

A She

B most

C **well-dressed**

D Party

Correct Answer : C

Q-48 Which part of the sentence has an error?
"I was shocked to see how my brother's behaviours has changed."

A I was shocked

B to see

C how my brother's

D **behaviours has changed**

Correct Answer : D

Q-49 Which sentence is grammatically correct?

A Neither of them are available.

B **Neither of them is available.**

C Neither of them were available.

D Neither of them have been available.

Correct Answer : B

Q-50 Identify the sentence with the correct verb form.

A If she was knowing the truth, she would have told us.

B Neither of the players were ready for the match.

C I wish I was invite to the meeting.

D **He insisted that she be present at the event.**

Correct Answer : D